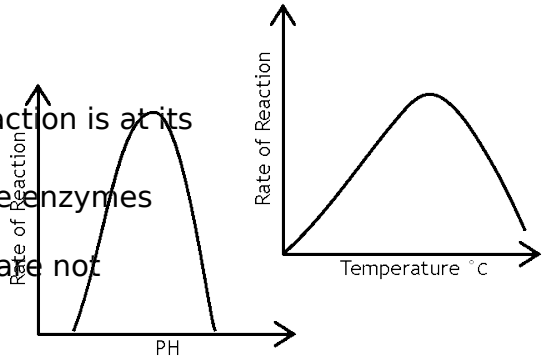
Enzymes are:

- Protein molecules
- Biological catalysts and increase the rate of chemical reactions
- Affected by temperature and pH.
- Found in animals, plants and microorganisms.
- They break large molecules into smaller molecules.
- They have a specific shape to fit a particular substrate.

Factors affecting enzyme action

1. Temperature

Enzymes work faster as the temperature increases.
They have an optimum temperature at which the rate of reaction is at its fastest.
At higher temperatures the rate of reaction decreases as the enzymes become denatured.
This is because the shape of the active site is lost and they are not able to form an enzyme-substrate complex.



2. pH

Changes in pH affect the shape of the active site.
Enzymes have an optimum pH.

Enzymes in digestion

Enzymes break down large insoluble food molecules into smaller soluble ones which can be absorbed through the small intestine wall to enter the bloodstream.

Substrate	Enzyme	Product	Where enzyme is produced	Optimum pH
Starch	amylase (a carbohydrase)	glucose	salivary glands & pancreas	7
Protein	trypsin (a protease)	amino acids	stomach, pancreas, & small intestine	3

Enzymes revision

Product	Enzymes	Explanation
Biological washing powder	lipases and proteases	Digest biological stains off clothes
Baby food	protease	Pre-digest protein to make digestion easier for babies
Slimming foods	isomerase (carbohydrase)	Converts glucose into fructose (less of which is needed to sweeten food)

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Lipids	lipases (fats)	fatty acids & glycerol	pancreas	8	

Enzymes are:

Factors affecting enzyme action

Enzymes in digestion

Substrate	Enzyme	Product	Where enzyme is produced	Optimum pH

Commercial use of enzymes

Product	Enzymes	Explanation

Teaching notes

Use as an information mat or poster when students are practising exam questions.

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Have one copy in the middle of the room, put the students in groups and ask them to fill in a blank copy of the sheet by taking it in turns to visit the mat and bring back one piece of information.

It could be used as a revision exercise for students to fill in themselves.

Differentiation

Make differentiated versions of the sheet by filling in some of the information. For extra challenge give students a version with errors in it and ask them to proofread it and then fill in a blank sheet with the corrections.

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